

Received 7 November 2025, accepted 22 December 2025, date of publication 2 January 2026,
date of current version 7 January 2026.

Digital Object Identifier 10.1109/ACCESS.2025.3649743

RESEARCH ARTICLE

UQA-RAG: Enhancing Urdu Question-Answer Retrieval With Semantic Similarity and LLMs

TALHA FAROOQ KHAN¹, MUHAMMAD ARSLAN², FAHAD ALTURISE³,
AHMED H. ALMUTAIRI⁴, AND MUHAMMAD TARIQ RAZA⁵

¹Department of Computer Science, The University of Faisalabad, Faisalabad 38000, Pakistan

²Department of Computer Science, University of Southern Punjab, Multan 60000, Pakistan

³Department of Cybersecurity, College of Computer, Qassim University, Buraydah 52379, Saudi Arabia

⁴Department of Computer Science, Northern Border University, Arar 73213, Saudi Arabia

⁵Department of Computer Science, Bahauddin Zakariya University, Multan 60000, Pakistan

Corresponding author: Fahad Alturise (Falturise@qu.edu.sa)

The Researchers would like to thank the Deanship of Graduate Studies and Scientific Research at Qassim University for financial support (QU-APC-2026).

ABSTRACT Urdu Question Answering (QA) systems struggle with limited annotated resources and linguistic complexities. These are significant hurdles for traditional Large Language Models (LLMs) that rely on static, pre-trained knowledge. This paper introduces UQA-RAG, a new framework to enhance Urdu QA by using Retrieval-Augmented Generation (RAG) over the multi-domain UNQA dataset. Our core methodological contribution is a Hybrid Retrieval Strategy. This approach overcomes the individual weaknesses of pure semantic or lexical search by combining their strengths. It fuses results from OpenAI's text-embedding-3-small vectors (Semantic) with BM25 (Lexical) using the Reciprocal Rank Fusion (RRF) algorithm. This method maximizes both context relevance and diversity. We conducted an extensive ablation study to evaluate UQA-RAG's performance with different LLM backbones, including GPT-4o, Llama-3.2, and Deepseek-r1. The optimal setup, pairing Hybrid Retrieval with GPT-4o, achieved a BERTScore F1 of 93.15%. This was a 16.55 percentage point improvement over our fine-tuned mBERT baseline. The system also reached a high RAGAS Context Precision of 95.20%. These results confirm that our hybrid approach is essential for generating accurate, contextually grounded responses, making UQA-RAG an effective solution for advancing high-quality QA in Urdu.

INDEX TERMS Large language models (LLMs), retrieval-augmented generation (RAG), question answering (QA), GPT-4o, Llama-3.2 (3B), Deepseek-r1.

I. INTRODUCTION

The recent years have witnessed the remarkable progress of natural language processing (NLP) that contributed to the enhancement of the creation of intelligent systems that comprehend and produce human language, especially Question Answering (QA) systems [1], [2]. Although QA systems have been very successful in high-resource languages such as English, they fail in low-resource languages, such as Urdu, because of absence of large-scale datasets and linguistic complexities. Urdu, the national language of Pakistan, is a major cultural and linguistic challenge in NLP and therefore a poorly explored domain of robust QA systems.

The associate editor coordinating the review of this manuscript and approving it for publication was Mu-Yen Chen¹.

The majority of QA systems are available only in high-resource languages due to access to large labeled datasets and large language models [3]. Nevertheless, Urdu presents some serious challenges, such as the absence of high-quality, large-scale datasets [4] and publicly available, domain-specific knowledge to train accurate models. In contrast to English, where large volumes of text data are available on the web, in books, etc., Urdu does not have such large-scale resources, especially in such domain-specific areas as law enforcement [5]. Urdu has a distinct linguistic structure that includes a unique script and syntax, which poses a challenge to the vast majority of traditional NLP tools and pre-trained models [6]. This has led to an urgent demand of effective, stable QA tools in Urdu-speaking communities, particularly in areas like law enforcement, government

services and education where it is necessary to give fast, accurate and contextually relevant responses.

The common practice of traditional QA techniques has been to use large pre-trained models and fine-tune them using domain-specific data [7]. Although it works well with high-resource languages, this approach would not scale to low-resource languages such as Urdu, as it requires large amounts of labeled data and a considerable amount of computational resources. Moreover, these systems are based on fixed pre-trained knowledge and this reduces their capacity to give real-time updates, a very important feature in dynamic fields such as law enforcement.

The Retrieval-Augmented Generation (RAG) can provide the practical answer to such problems because it combines information retrieval and text generation into one. This hybrid model enables the model to obtain and make use of the modern information stored in other knowledge repositories in the production of responses. There are two major benefits of this structure two-fold: in addition to the ability to retrieve information in real-time, it generates text with a high level of power, and less relies on huge training collections [8]. This architectural freedom makes RAG especially suitable to the language such as Urdu because it can be more accurate with the help of certain external sources, including the ones provided by the government and the records belonging to people.

In this paper, UQA-RAG has been introduced and it is a Retrieval-Augmented Generation (RAG) model that can be utilized to improve the outcome of Urdu QA tasks. The suggested system operates on a dataset that is a custom built set of data that builds on three leading Urdu news media: Jang, express and Dawn. The dataset is referred to as Urdu News QA (UNQA) and includes more than 900 articles and 3,600+ hand prepared QA pairs. The articles are divided into six separate fields that include Politics, Business and Economics, Sports, Science and Technology, Health and Entertainment giving a rich general purpose knowledge base. The RAG offers an opportunity to unite the two worlds i.e. information retrieval and generative modeling to generate very relevant and context sensitive answers to the queries posted by the users in Urdu.

This paper makes two key contributions:

- a) The development of a new modified RAG system, called UQA-RAG, specifically designed for Urdu Question Answering tasks. This system integrates an enhanced information retrieval framework with a large language model to produce highly relevant and context-aware responses in a low-resource language.
- b) The introduction of a novel hybrid retrieval strategy that combines semantic similarity search with a keyword-based retrieval algorithm (BM25). This approach, enhanced by Reciprocal Rank Fusion (RRF), is a significant methodological contribution that improves the accuracy and effectiveness of information retrieval by leveraging the strengths of both dense and sparse retrieval methods.

The first of our contributions concerns the problem of low novelty of standard RAG systems with the suggestion of a hybrid retrieval strategy, UQA-RAG. This plan is more efficient than one method of retrieval, which produces more precise outcomes. Its efficiency is the most vital trait: the systems embedded within the system compute the embeddings and depend on the prompt retrieval system to match the queries and give the highly relevant answers at the lowest possible cost in terms of computation. This makes our model to be scaleable without losing the quality of answers.

The second contribution is a new hybrid retrieval methodology that will solve the limitations of one-way retrieval methodology. Although semantic search is the best when it comes to capturing meaning and context of a query, it is at times deficient in terms of detecting documents that have literal matches of a keyword. On the other hand, the keyword-based methods are quick and accurate when there is a literal search but do not comprehend the meaning of a query. We use a combination of these two techniques and combine their output intelligently with Reciprocal Rank Fusion (RRF), in such a way that our system is capable of finding the most useful information to a greater number of user queries, which is much more performance-effective and reliable.

II. LITERATURE REVIEW

A. ADVANCEMENTS IN LARGE LANGUAGE MODELS (LLMs)

LLM has developed rapidly and transformed NLP since it can understand and write human-like language. The fundamental architecture of most of the most popular models, including OpenAI GPT series, Google PaLM, and Meta LLAMA series, is Transformer architecture. Its architecture can process sequential data (e.g. text) with a sophisticated attention mechanism, allowing the model to consider the importance of words in the input sequence, and finding complex dependencies in language [9], [10]. The Transformer has proven effective, and the use of LLMs has become commonplace because of their exemplary performance in the areas of text generation, translation, and summarization.

Transformer architectures are found in many forms. Decoder only models, including GPT series, are primarily used in text generation where encoder only models, including BERT and RoBERTa models, are more applicable in classification and named entity recognition. The third, the encoder-decoder model like BART, is a general model that can be trained to do both generation and understanding tasks [11], [12]. This architectural diversity enables the specialisation of the LLMs in complex linguistic tasks.

Question Answering (QA) is one of the most visible research areas that use LLMs. The problem however is that a major challenge comes in when LLMs which are trained on general datasets need to respond to questions in specific fields such as law, medicine or education [13], [14], [15], [16]. The human-like response generation capabilities of LLMs face challenges when dealing with specialized knowledge because

their training occurs through general datasets [17]. The occurrence of hallucination emerges when models produce answers that contradict factual evidence or the available data which represents a significant drawback of LLMs [18].

To address this significant limitation, the Retrieval-Augmented Generation (RAG) approach has become an interesting alternative. RAG combines the strength of generative models with external document retrieval systems. The basic concept, originally suggested by Lewis et al. in 2020 [19], is to access a knowledge base and feed the relevant information to the LLM prior to it generating a response [20], [21]. The method will make the answer of the model based on verified and current external knowledge instead of the internal and fixed training data. Further advancements like Probably Asked Questions (PAQ) and QA-pair retriever RePAQ have also optimized this process by giving the model structured knowledge to better retrieve relevant information to a particular query [22].

B. RAG IN SPECIALIZED DOMAINS AND LOW-RESOURCE LANGUAGES

RAG has been applied to many specialized areas on a large scale. In the legal field, where factual accuracy is most important, such tools as LRAGE have been designed to specifically test RAG systems, and the quality of retrieval components is stressed [23]. On the same note, the medical and biomedical disciplines have adopted RAG to improve question-answering systems and clinical decision support [24]. This demonstrates the usefulness of RAG in areas where precision is not an option.

One of the most fundamental problems, especially with low-resource languages, in terms of creating effective QA systems is that such datasets are not extensive and high-quality. In the case of languages such as Urdu, much of the research has been dedicated towards producing and developing such resources. As an example, [25] created UQuAD1.0, a large-scale Urdu dataset on machine reading comprehension, and [26] created the UQA dataset, covering the problems of answer span alignment in English and Urdu because of the grammatical differences between the languages. More recently, [27] was concerned with the necessity of domain specific resources and introduced LEGAL-UQA, a parallel Urdu-English legal QA dataset. A detailed survey by [28] gives a taxonomy of RAG architectures, which is essential in the awareness of the diversity of implementations in the field.

C. CHALLENGES IN RETRIEVAL-AUGMENTED GENERATION (RAG) FOR QA SYSTEMS

The successful application of RAG depends heavily on the effectiveness of its external document retrieval component. A primary challenge is the need to retrieve documents that are not only relevant but also accurate and current [29], [30], [31], [32]. While traditional Open-Domain Question Answering (ODQA) often relies on publicly available datasets, the growing demand for specialized, domain-specific datasets requires

TABLE 1. Tabular overview of related QA systems.

Reference	Domain / Language	Methodology	Key Contribution
UQuAD1.0 (Khoja & Ahmed, 2021) [25]	Open-Domain / Urdu	Crowdsourced & Machine-translated SQuAD	Creation of a large-scale Urdu dataset for extractive QA.
UQA (Arif et al., 2024) [26]	Open-Domain / Urdu	SQuAD translation with EATS technique	Addresses the challenge of answer span alignment in translated text.
LEGAL-UQA (Faisal & Yousaf, 2024) [27]	Legal / Urdu-English	GPT-4-assisted generation & manual refinement	The first legal QA dataset for Urdu.
Lewis et al. (2020) RAG [19]	General / Knowledge-Intensive	Retriever-Generator Architecture	The foundational RAG model, combining a retriever with a pre-trained generator.
Mahmood et al. (2025) [34]	Medical / Urdu	Comparative study of fine-tuning, CA-Gen, and RAG	Demonstrates RAG's effectiveness in a low-resource medical domain.
LRAGE (Kang et al., 2023) [23]	Legal / General	Evaluation framework for RAG systems	Highlights the importance of domain-specific RAG components.

custom web retrieval, content extraction, and segmentation capabilities to build relevant knowledge bases.

Web retrieval and content extraction processes are usually time-consuming and complicated and need a thorough knowledge of the domain in order to find and extract the most suitable content. A risk of hallucination also exists in RAG systems when the documents retrieved in the corpus are irrelevant and/or conflicting [33]. This issue is even magnified when it comes to unstructured or noisy web-based data sources. As a result, the quality of the retrieved information and its relevance to the query are critical to a sound RAG system.

III. METHODOLOGY

Our research methodology employs a two-pronged approach. First, we establish a comparative baseline using a traditional extractive Question-Answering (QA) model. Second, we develop the UQA-RAG system, which combines a retrieval framework with modern language models. We designed this dual structure to demonstrate the RAG approach's effectiveness by directly comparing its performance against a non-RAG baseline.

A. BASELINE EXTRACTIVE QA MODEL

We determined a quantitative baseline upon which we evaluated the performance of our system. To this end, we adjusted

a multilingual BERT (mBERT) model, though we decided to use it to reflect the traditional extractive QA systems. Such models operate based on the identification and extraction of the most suitable span of answers within a certain setting. In order to make a reasonable comparison, we conditioned this mBERT model on the very same Urdu QA dataset on which UQA-RAG was conditioned to enable direct comparison of scores. This baseline is a good point to show the performance improvement that our proposed UQA-RAG framework has brought.

B. QUESTION-ANSWER EXTRACTION FROM A LIMITED CORPUS

The problem here is to recover the most suitable answer to a confined document set and natural language query via the integration of contextual embeddings and similarity search approaches. The system scans documents that contain multiple sentences of an objective of identifying the best response without using outside sources.

i. Problem Definition:

- a. Given a set of documents $D = \{d_1, d_2, d_3, \dots, d_n\}$ and a user query Q , the objective is to identify the most relevant answer A^* within D , ensuring maximum semantic alignment with Q .

ii. Optimization Approach:

The retrieval process is formulated as:

$$A^* = \operatorname{argmax}_A S(Q, A) \quad A \in D$$

where $S(Q, A)$ represents a scoring function that quantifies the semantic similarity between query Q and candidate answers A . This function is computed using contextual embeddings, cosine similarity, and language model evaluations.

The retrieval system is designed to maximize (Q, A) , ensuring that the selected answer A^* is the most contextually relevant response to Q .

C. DATA COLLECTION AND PREPROCESSING

- i. Data Source: We constructed the dataset by scraping articles from three of Pakistan's most prominent news websites: Jang News, Express News, and Dawn News Urdu. To ensure the dataset was diverse and broadly relevant for NLP tasks, we collected articles from six distinct categories: Politics (سیاست), Business & Economics (کاروبار), Sports (کھیل), Science & Technology (سائنس و ٹیکنالوجی), Health (صحت), and Entertainment (تفریح).
- ii. Preprocessing Steps:
 - Article Collection: Approximately 900 unique articles were gathered from the specified sources.
 - Question-Answer (QA) Pair Generation: These articles had 3,600 QA pairs that were created manually. This was a manual procedure that produced a high-quality dataset with a broad focus of topics. We had included some simple factual questions

and some complex analytical questions in order to enable us come up with an exhaustive assessment of the same.

- Text Cleaning and Delimitation: The text was carefully purged of all unnecessary HTML code, advertisements and other noise. Manually, triple backticks (```) were added to the articles to standardize text parsing, and to make the programmatic segmentation of the semantically coherent chunks consistent in formatting.
- Final Dataset Compilation: The QA pairs were integrated with their respective contexts, forming a structured corpus suitable for training and evaluation.

D. THE UQA-RAG FRAMEWORK

The UQA-RAG system that we are proposing is a multi-phase system that would involve both information retrieval and text generation. Such a framework is constructed to recall the most suitable context to a certain query to the natural language and apply it to produce a specific and contextually relevant response. The architecture of the system can be considered in two major sections: Hybrid Retrieval and Response Generation.

- i. **Hybrid Retrieval Strategy:** Our UQA-RAG system involves the use of a new hybrid framework to achieve a high level of retrieval accuracy. It is a way of expanding upon the scope of one search technique to create a hybrid of semantic and lexical (keywords) search. A hybrid approach is much stronger, which we discovered. Semantic search is good at comprehending the purpose and meaning of a query whereas the keyword search is not good at locating certain names, words, or phrases. We have a system that exploits the benefits of both. The results of these two complementary methods are combined to give the final relevance score of each document.

- Semantic Search & Embedding Storage: To search the textual data in our semantic search case, we initially transformed the textual data into meaningful vectors with the text-embedding-3-small and text-embedding-3-large text-embedding models of openAI. In order to make retrieval efficient and scalable we incorporated ChromaDB as our high-dimensional vector database. In ChromaDB, the contextual embeddings of text-embedding-3-small were indexed, and each one of the vectors was associated with the corresponding QA pair ID. On a querying command by the user, we dynamically compute the embedding of the query before retrieving the best matches in the database using cosine similarity. We have tried various retrieval settings: Top-1 Retrieval that gives preference to just one most relevant document and Top-3/Top-5 Retrieval which increases the search range.

This increased coverage has enabled us to include numerous helpful documents, a technique that we have discovered to improve the diversity of the context and the integrity of the resulting response. Cosine similarity between a query and a document vector is given as:

$$\text{sim}_{\cos}(Q, D) = \frac{Q \cdot D}{\|Q\| \|D\|}$$

For reproducibility, the text-embedding-3-small model (1536-dimensional embeddings) was used for all reported experiments, as it showed superior performance in Top-3 and Top-5 retrieval scenarios. All embeddings were L2-normalized before indexing and similarity computation. Context chunks were created with an average length of 300–400 tokens. During retrieval, a fixed Top-5 semantic candidates were selected based on cosine similarity, without applying a hard similarity threshold, to avoid excluding relevant context in a low-resource setting.

- **Keyword-Based Search Component:** We implemented the BM25 (Best Match 25) ranking function to perform keyword-based retrieval. BM25 is highly effective at finding documents with exact keyword matches, which is crucial for specific, factual queries. The BM25 score for a document d given a query q is calculated as:

$$\text{BM25}(d, q) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, d) \cdot (k_1 + 1)}{f(q_i, d) + k_1 \cdot (1 - b + b \cdot \frac{d}{\text{avgdl}})}$$

Here, **IDF** is the Inverse Document Frequency of the query term, $f(q_i, d)$ is the term frequency of the query term in the document, $|d|$ is the document length, **avgdl** is the average document length.

The BM25 hyperparameters were fixed to standard values ($k_1 = 1.5, b = 0.75$) across all experiments. Urdu-specific tokenization and stopword removal were applied prior to indexing to improve lexical matching accuracy and reduce noise.

- **Reciprocal Rank Fusion (RRF):** After retrieving ranked lists from both the semantic and keyword-based search components, we merge them using Reciprocal Rank Fusion (RRF). This method is a state-of-the-art rank fusion algorithm that assigns a final score to each document based on its rank in both lists. The RRF score for a document d is calculated as:

$$\text{RRFScore}(d) = \sum_{l \in \{s, k\}} \frac{1}{k + \text{rankl}(d)}$$

where s and k represent the semantic and keyword lists, $\text{rankl}(d)$ is the rank of the document in list l , and k is a constant.

In our implementation, the RRF constant was set to $k = 60$, and the Top-5 candidates from both semantic and BM25 retrieval were fused to form the final ranked list. This configuration was kept fixed across all experimental settings to ensure consistent and reproducible retrieval behavior. From the RRF-fused ranking, the Top-3 context chunks were selected and concatenated to form the final context provided to the language model during answer generation. All retrieval-related configurations, including embedding model choice, normalization strategy, retrieval depth, BM25 parameters, and RRF fusion settings, were held constant across experiments to enable full reproducibility and fair comparison of results.

- ii. **Response Generation using Large Language Models** Once the most contextually relevant snippets have been identified, advanced Large Language Models (LLMs) were used to provide accurate and contextually relevant responses. The top- k text snippets retrieved were concatenated with the original query of the user to form a rich context-aware prompt. The prompt template was designed in such a way that it is clear that the LLM should develop its answer solely on the given context, thus reducing the chances of hallucination. The usual prompt format was the retrieved context then the query with explicit instructions like: “Answer the question in your own words based on the given context. When the answer is not in context, then answer by saying, the information is not available.” The models tested were GPT-4o, a closed-source model with high-quality text generation and contextual reasoning; Llama-3.2 (3B), an efficient open-source model that focuses on retrieval-augmented generation tasks; and Deepseek-r1 (7B), a larger-scale open-source model that was used to do comparative testing. The Top-3 context snippets that were relevant to each model were used as an input to generate answers, and the resulting answers were systematically tested with automated metrics to measure coherence, accuracy, and semantic alignment with the original question.

IV. EXPERIMENTAL SETUP AND EVALUATION

A. DATA PREPARATION

Data preparation stands as an essential requirement for machine learning and NLP-based tasks, especially during the development of Retrieval-Augmented Generation (RAG) systems. This study utilized a significantly larger and more diverse corpus, the Urdu News QA (UNQA) dataset, which was collected through web scraping from three of Pakistan’s most reputable news sources: Jang News, Express News, and Dawn News Urdu. The raw data was collected from 900 unique articles systematically categorized into six distinct domains Politics, Business & Economics, Sports, Science & Technology, Health, and Entertainment. The UNQA dataset contains 3,600 manually curated question-answer

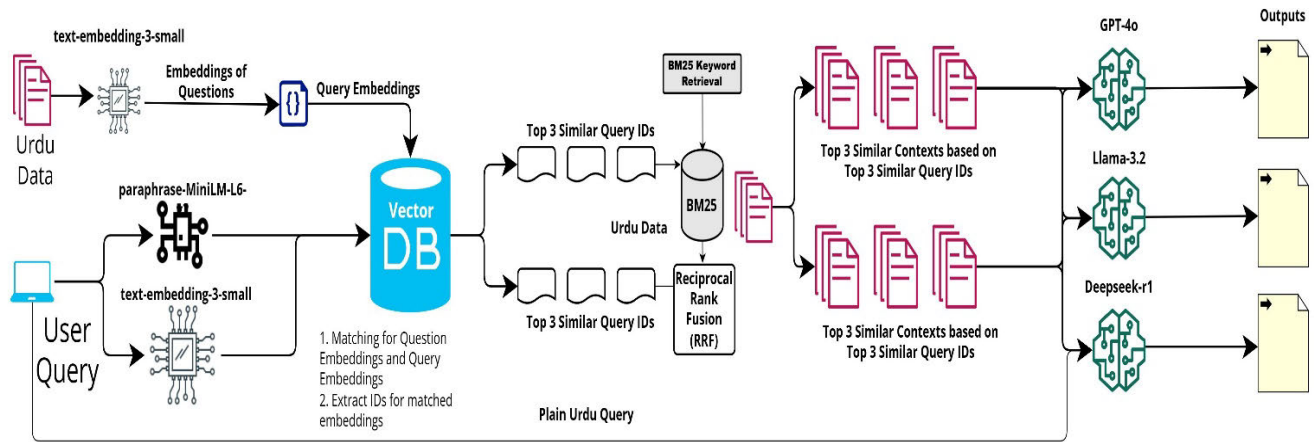


FIGURE 1. UQA-RAG system architecture featuring hybrid retrieval (Semantic and BM25) fused by RRF.

pairs, providing a robust and general-purpose knowledge base, in contrast to the single-domain focus of older datasets. The raw HTML content was passed through a text content extraction process to strip out extraneous scripts, styles, and other undesired elements, making the content suitable for machine learning applications. The preprocessing pipeline began by the removal of unwanted characters and the minimization of extra white spaces, followed by Urdu stopword removal to eliminate unimportant content and normalization of words to address various forms and spellings of words. Metadata tagging of the RAG process allowed the system to have improved contextual retrieval capability, dividing articles into sections. The formatted QA pairs are usable in retrieval-based applications following the data preparation process.

B. GROUND TRUTH PREPARATION

The ground truth preparation process must be subject to human verification to come up with a correct dataset that can be used as the benchmark to evaluate the model output. The ground truth is composed of 3,600+ QA pairs that include questions and their corresponding answers drawn out of multi-domain articles from the collected news sources. The QA pairs were reviewed and validated manually by experts in the field to ensure their factual correctness and semantic relevance. This process involved several steps:

1. **Selection of Relevant Questions:** Questions were selected based on their relevance to the articles. Each question was designed to probe different aspects of the news domains, including policy, economic trends, sports results, and scientific developments.
2. **Answer Validation:** For each selected question, the corresponding answer was verified against the original articles to ensure that the answer was accurate, complete, and aligned with the information available in the source material.

3. **Manual Labeling:** To prepare the dataset for evaluation, each QA pair was labeled with metadata (e.g., difficulty level, topic of inquiry) to make the evaluation process more comprehensive.
4. **Consistency Check:** A final step was taken to ensure that answers were consistent across different QA pairs related to similar topics. This minimizes biases and ensures that the generated answers are contextually relevant.

The ground truth is used as a reference point to the performance of the RAG system. It is also important in informing the development of the retrieval and generation systems through the provision of the expected outputs of different inputs.

C. HYBRID RETRIEVAL MECHANISM CONSTRUCTION

An inverted index and vector index serve as efficient data structures which connect terms (question embeddings) to their associated document or QA pair. The retrieval of relevant content from large datasets requires an efficient and fast mechanism [35]. The system constructs its retrieval architecture through semantic embeddings that represent questions (for vector search) and keyword indexing (for lexical search) for improved matching accuracy.

1) EMBEDDING REPRESENTATION

Each question in the dataset is transformed into a dense vector representation using an embedding model. OpenAI's text-embedding-3-small model is employed to generate high-quality semantic embeddings for all context chunks in the UNQA dataset. The embedding process can be mathematically represented as:

$$q_i = \text{Embed}(Q_i)$$

where Q_i is the i -th question, and q_i is the embedding of question.

2) VECTOR DATABASE

The embeddings of all context chunks are stored in a vector database, such as ChromaDB. This allows the system to retrieve relevant context based on the similarity of the query embedding to the stored context embeddings.

3) COSINE SIMILARITY

Once a query embedding q_q is generated, the cosine similarity between the query embedding and all stored context embeddings is computed to identify the most relevant context. The cosine similarity (q_q, q_i) between the query q_q and each question embedding q_i is calculated as follow:

$$\text{Cosine Similarity}(q_q, q_i) = \frac{q_q \cdot q_i}{\|q_q\| \|q_i\|}$$

The query embedding q represents the vectorized form of the input query, while q_i denotes the embedding of the i -th context chunk in the dataset. The magnitudes $\|q_q\|$ and $\|q_i\|$ correspond to the norms of the query and context embeddings, respectively. Cosine similarity is computed between q and all stored context embeddings to determine relevance, and the top-N context chunks with the highest similarity scores are retrieved for further processing.

D. RETRIEVAL AND RANK FUSION

The retrieval process functions as the central component of the RAG system by finding the most suitable context chunks that match a specified query. The system starts by embedding user queries through the same model used for index construction, which includes text-embedding-3-small. This model generates dense vectors with a dimensionality of 1536.

1. Dual Retrieval Execution: The system executes two parallel searches:

- Vector Search: Calculates cosine similarity against the ChromaDB index for semantic relevance.
- Lexical Search: Executes the BM25 algorithm against a traditional inverted index for keyword relevance. The BM25 parameters are set to $k_1 = 1.2$ and $b = 0.75$ to optimize for the unique script and document lengths in the Urdu corpus.

2. Reciprocal Rank Fusion (RRF): The ranked lists from both the Semantic and Lexical searches are merged using RRF to create a single, highly relevant list of context chunks. This mechanism can be represented as:

$$\text{RRFScore}(d) = \sum_{l \in \{s, k\}} \frac{1}{k \cdot \text{rank}_l(d)}$$

where s and k represent the semantic and keyword lists, $\text{rank}_l(d)$ is the rank of the document in list l , and k is a constant, typically set to 60.

3. Final Selection: The system retrieves the top-N most similar and relevant context chunks from the RRF-fused index according to their calculated similarity scores to ensure relevant content selection. The

retrieval mechanism can be represented as:

$$D_{\text{Retrieved}} = \text{TopN}_{\text{RRF}(\text{search}_{\text{Vector}}(Q) \cup \text{Search}_{\text{BM25}}(Q))}$$

where N is the number of retrieved context chunks (typically set to 3 to 5 for generation). The result is a set of contextually relevant chunks that are used as input to the generation model to produce a response. Figure 1 illustrates the retrieval pipeline of UQA-RAG, where user queries and stored context embeddings are processed through a Hybrid Retrieval system to retrieve the most contextually relevant context.

E. GENERATION

The system moves to the generation phase to create a response using GPT-4o, Llama-3.2, or Deepseek-r1 after retrieving the top-N relevant context chunks. The generative model receives the retrieved context during contextualization before processing this information. The model receives enough background information through this process to create responses that maintain contextual accuracy and semantic precision. The language model proceeds to the response generation step by processing the input to create an answer while maintaining coherence and factual accuracy and linguistic fluency while staying close to the ground truth [36]. The mathematical representation of response generation appears as follows:

$$R_{\text{generated}} = \text{LLM}(Q, D_{\text{Retrieved}})$$

where $R_{\text{generated}}$ represents the generated response and $D_{\text{Retrieved}}$ denotes the context derived from the top-N retrieved context chunks. This structured approach enhances the model's ability to produce precise and contextually enriched answers.

F. COMPARISON TO TRADITIONAL RAG SYSTEM

Conventional Retrieval-Augmented Generation (RAG) systems combine the retrieval and generation processes in a single processing pipeline with a single model retrieving documents and generating responses. These systems use traditional methods of retrieval like BM 25 and TF- IDF to locate pertinent documents however, these methods concentrate on lexical similarities rather than semantic similarity. The UQA-RAG system has overcome this limitation by the use of Hybrid Retrieval strategy.

The suggested RAG system breaks down its operation into three distinct steps in order to maximize the process of retrieval and generation. The main distinction is in the retrieval stage involving dense vector search and lexical search and employing RRF to address the shortcomings of each technique. In retrieval, the system converts both documents and queries to dense vectors (Semantic Search) and at the same time performs a keyword-based search (BM25) and identifies the nearest matches with Reciprocal Rank Fusion (RRF). The system is more accurate in retrieving the results as it is semantically aware and maintains the precise matches of keywords. Once the context has been located, the system generates responses using GPT-4o, Llama-3.2, or Deepseek-r1.

The split between retrieval and generation process improves the accuracy of the responses and also assists the model to generate text that perfectly fits in the context. The system design enables us to update the individual components of the system which enables us to add new retrieval and generation technologies without significant alterations.

G. EVALUATION METRICS

To measure the efficiency of the suggested Retrieval-Augmented Generation (RAG) system and the fine-tuned mBERT baseline model, several evaluation measures are used, both at the retrieval accuracy and the quality of the generated texts.

1) BERTSCORE

Semantic comparison of the automatic response and reference answers is quantified using BERTScore that outperforms the traditional word-matching evaluation techniques. The similarity of the generated response with reference answers [37] is also determined by using BERTScore to compute the cosine similarity between word embeddings. It is calculated as:

$$BERTScore(r, r') = \frac{1}{|r|} \sum_{i=1}^r \max_{j=1}^r \text{cosine_similarity}(r_i, r'_j)$$

where r_i and r'_j are the word embeddings of the words in the generated and reference answers, respectively.

2) RETRIEVAL-AUGMENTED GENERATION ASSESSMENT

Retrieval-Augmented Generation Assessment (RAGAS) is a holistic measure that measures both retrieval and generation process of the system [38]. The metrics of this group are: Faithfulness, Answer Relevance, Context Precision, and Context Recall. RAGAS score is shaped by weighted combination of retrieval and generation quality:

$$\begin{aligned} RAGAS \text{ Score} = & w_1. \text{Faithfulness} \\ & + w_2. \text{Answer Relevance} \\ & + w_3. \text{Context Quality} \end{aligned}$$

where w_i is a user-defined weight controlling the balance between retrieval precision and response generation quality.

3) TOP-N RETRIEVAL ACCURACY

The retrieval phase assessment measure is used to ascertain the ability of the system to retrieve the most suitable questions among the existing datasets. When effectively retrieving the most contextually appropriate information, the system is said to have good retrieval capabilities:

$$Top - N \text{ Retrieval Accuracy} = \frac{\sum_{i=1}^N 1(q_i \in R)}{N}$$

4) RESPONSE SIMILARITY

Response similarity is used to determine the cosine similarity between the response generated and the reference answer

to maintain consistency in the system to give coherent and contextually correct answers. It is computed as:

$$\text{Response Similarity} = \frac{G \cdot R}{\|G\| \|R\|}$$

where G is the embedding vector of the response that has been generated and R is the embedding vector of the reference answer. The denominator is the magnitudes of these embeddings and the numerator the alignment of these embeddings in semantic space as the dot product. As the similarity score increases, this means that the response that has been generated is very similar to the reference response in meaning and relevance. This measure is applied to the RAG system alone, since the extractive mBERT baseline model does not produce a response, but it extracts it out of the context.

5) EXACT MATCH (EM)

The Exact Match (EM) score is a stringent, binary metric used to determine the factual precision of the generated response. It assesses the model's ability to provide a complete and precise factual answer by requiring a perfect match between the generated answer (A_{gen}) and one of the ground-truth reference answers (A_{ref}) after normalization.

$$EM = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\text{Normalize}(A_{gen}^{(i)}) = \text{Normalize}(A_{ref}^{(i)}))$$

where N is the total number of questions, and $\mathbb{I}(\cdot)$ is the indicator function that equals 1 if the condition is true and 0 otherwise.

6) F1 SCORE (TOKEN-BASED)

The F1 Score (Token-Based) measures the average token overlap between the generated answer and the ground-truth answer. This metric is the traditional F1 score used in Question Answering and is calculated based on the harmonic mean of token-level precision (P) and recall (R), representing the model's extraction and generation quality.

$$\begin{aligned} P &= \frac{\text{Number of overlapping tokens}}{\text{Total tokens in } A_{gen}} \\ R &= \frac{\text{Number of overlapping tokens}}{\text{Total tokens in } A_{ref}} \\ F1_{Token} &= 2 \cdot \frac{P \cdot R}{P + R} \end{aligned}$$

V. RESULTS AND DISCUSSION

The study conducted extensive experiments on the Urdu News QA (UNQA) dataset to enhance both the retrieval precision and response of an Urdu Question Answering system with the integration of Retrieval-Augmented Generation (RAG) strategies and state-of-the-art embedding models as well as Large Language Models. Findings of the experiment are structured into three main parts, an embedding comparison between purely semantic retrieval and Comparative Analysis of the new Hybrid Retrieval Strategy (Semantic + BM25 + RRF) with different RAG settings, and a

strict quantitative segment comparison between the optimized UQA-RAG system and a traditional extractive mBERT baseline model with an entire set of measures, such as Exact Match (EM) and F1 Score (Token-Based).

A. EMBEDDING COMPARISON (PURE SEMANTIC RETRIEVAL)

In this section the text-embedding-3-small and text-embedding-3-large models of OpenAI embedding were evaluated by calculating cosine similarities between query and context embeddings in a pure vector search environment. Evaluation was carried out through three types of retrieval; Top-1, Top-3 and Top-5 retrieval. The study objective was to identify the model that was most suitable in recovering the relevant answers using similarity score.

Top-1 Retrieval:

- text-embedding-3-small: Mean Similarity = 0.59%
- text-embedding-3-large: Mean Similarity = 0.62%

Text-embedding-3-large model had a slightly better performance in Top-1 retrieval compared to the small model by generating a mean similarity of 0.62% compared to 0.59%. The bigger model proves to be more efficient in choosing the best single outcome that may be critical in applications that require the best solutions.

Top-3 Retrieval:

- text-embedding-3-small: Mean Similarity = 0.72%
- text-embedding-3-large: Mean Similarity = 0.70%

A much better result was the mean similarity score of 0.72% of the Top-3 retrieval of the text-embedding-3-small model as opposed to 0.70% of the large model. The small model does slightly better on the retrieval of the three most relevant results as requested. The small model seems to have an advantage of offering better performance in search of diversity in relevant matches.

Top-5 Retrieval:

- text-embedding-3-small: Mean Similarity = 0.72%
- text-embedding-3-large: Mean Similarity = 0.67%

Again, in Top-5 retrieval, the text-embedding-3-small model was better in comparing with the text-embedding-3-large model with a mean similarity of 0.72 at their individual scores and 0.67, respectively. This supports the conclusion that the small model is more efficient in terms of its ability to retrieve several relevant matches and it is therefore the superior one when it comes to working with a wider range of results.

The text-embedding-3-small is better than the text-embedding-3-large since it gave better results on Top-3 and Top-5 retrieval tasks in these particular settings. The working model that was more efficient in Top-1 retrieval was the text-embedding-3-large model since it was able to retrieve the most relevant single match. The small and large models are based on the need of tasks as small model is best in broad result retrieval but the large model is best at single match retrieval.

B. COMPARATIVE ANALYSIS OF UQA-RAG AND FINE-TUNED MBERT BASELINE

The system created embeddings for user queries and context data. The query embeddings underwent comparison through two models including paraphrase-MiniLM-L6-v2 and text-embedding-3-small to support both the Pure Semantic and Hybrid Retrieval strategies. The three most relevant context results were chosen to undergo analysis through GPT-4o, Llama-3.2 (3B), and Deepseek-r1.

Exact Match (EM), F1 Score (Token-Based), BERTScore (precision, recall, F1 score), response similarity, and the RAGAS Context Precision measure were all used to comprehensively assess the quality of responses. Table 1 provides a performance comparison between various retrieval strategies and LLMs, and also shows the performance of a fine-tuned mBERT base line. The findings underline the fact that the Hybrid Retrieval Strategy combined with GPT-4o is the most stable in terms of the overall performance measures.

The results of the analysis of the quality of the response provided by three LLMs and the mBERT baseline showed significant discrepancies in the performance:

- Comparison to Baseline and Hybrid Validation: The UQA-RAG system consistently and significantly outperforms the traditional fine-tuned mBERT extractive baseline across all metrics. While the mBERT model achieved a BERTScore F1 of 76.60%, alongside a standard Exact Match (EM) score of 45.10% and a token-based F1 Score of 61.30%, the UQA-RAG system using the Hybrid Retrieval Strategy with GPT-4o demonstrated a massive gain. This optimal configuration reached a BERTScore F1 of 93.15%, an EM score of 78.45%, and an F1 Score (Token-Based) of 88.90%. This represents a superior 33.35 percentage point gain in factual precision (EM), demonstrating that the RAG framework, particularly with the novel Hybrid Retrieval approach, is a more effective method for handling the complexities of accurate, context-grounded Urdu question-answering than a purely extractive approach.
- GPT-4o Produced Highest Quality Responses: GPT-4o produced the highest quality responses across all configurations. When paired with the Hybrid Retrieval Strategy (text-embedding-3-small + BM25 + RRF), it achieved a BERTScore precision of 91.80%, recall of 94.57%, and an F1 score of 93.15%. This optimal configuration also delivered a high EM score of 78.45% and the highest RAGAS Context Precision at 95.20%, indicating that GPT-4o generated the most accurate, relevant, and contextually coherent answers using the highest quality context provided.
- Llama-3.2 (3B) Performance: Llama-3.2 (3B) also demonstrated strong performance, particularly when using paraphrase-MiniLM-L6-v2 for context embeddings. It achieved a high F1 Score of 82.10% and a 65.50% EM score. However, its BERTScore

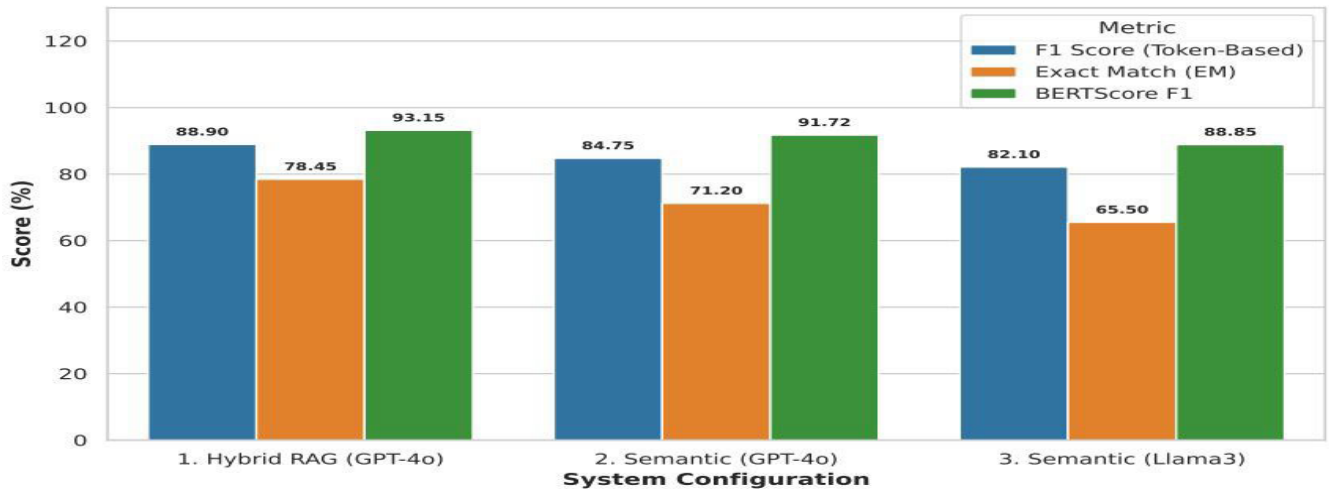


FIGURE 2. Comparison of the hybrid RAG architecture (GPT-4o) against the extractive mBERT Baseline on UQA metrics.

precision and recall were lower than GPT-4o (88.85% F1 score), suggesting that while it can produce semantically relevant responses, it was not as effective in generating highly precise and fully correct (Exact Match) responses.

- **Deepseek-r1 Grounding Issues:** Deepseek-r1, while performing reasonably well in terms of BERTScore context matching, yielded significantly lower response similarity scores. In the case of a combination of text-embedding-3-small and paraphrase-MiniLM-L6-v2, Deepseek-r1 yielded a response similarity of only 12.77%, which indicates that it is unable to produce contextually relevant answers in comparison with GPT-4o and Llama-3.2. This performance deficit is further reflected in its low Exact Match (EM) score (57.90% with text-embedding-3-small) and corresponding F1 Score (69.80%), confirming that the model fails to generate responses that are strictly and factually grounded in the provided context, even when that context is of high quality (RAGAS Context Precision of 81.45%).

When the retrieval strategy was changed to Pure Semantic Search using two text-embedding-3-small models (one for query, one for context), GPT-4o still performed well but experienced a significant decrease in overall precision. The model saw its Exact Match (EM) score drop by 7.5 percentage points (from 78.45% to 70.95%) and its F1 Score (Token-Based) drop by 4.4 percentage points (from 88.90% to 84.50%) in comparison to the Hybrid Retrieval. The performance of both Llama-3.2 and Deepseek-r1 also worsened across all metrics, which further proves the necessity to choose an appropriate retrieval method that is robust against the natural inconsistencies and low recall of pure vector search, thus validating the Hybrid Strategy.

GPT-4o demonstrates superior performance as the best model for producing high-quality responses, especially when

combined with our novel Hybrid Retrieval approach. The model delivered the most precise and contextually appropriate answers which resulted in its position as the most dependable solution for tasks demanding accurate responses. The Llama-3.2 (3B) model showed strong performance in response similarity but fell short of GPT-4o in precision and recall measurements. The responses from Deepseek-r1 were poor even though its context matching scores were acceptable.

When the retrieval strategy was changed to Pure Semantic Search using two text-embedding-3-small models (one for query, one for context), GPT-4o still performed well but experienced a significant decrease in overall precision. The model saw its Exact Match (EM) score drop by 7.5 percentage points (from 78.45% to 70.95%) and its F1 Score (Token-Based) drop by 4.4 percentage points (from 88.90% to 84.50%) in comparison to the Hybrid Retrieval. The performance of both Llama-3.2 and Deepseek-r1 also worsened across all metrics, which further proves the necessity to choose an appropriate retrieval method that is robust against the natural inconsistencies and low recall of pure vector search, thus validating the Hybrid Strategy.

GPT-4o demonstrates superior performance as the best model for producing high-quality responses, especially when combined with our novel Hybrid Retrieval approach. The model delivered the most precise and contextually appropriate answers which resulted in its position as the most dependable solution for tasks demanding accurate responses. The Llama-3.2 (3B) model showed strong performance in response similarity but fell short of GPT-4o in precision and recall measurements. The responses from Deepseek-r1 were poor even though its context matching scores were acceptable.

This work shows that the retrieval strategy and LLM choice are a crucial component of optimising Urdu QA systems based on Retrieval-Augmented Generation (RAG). The text-embedding-3-small model was better at Top-3 and

TABLE 2. Performance comparison of different query and context embeddings with various LLMs and mBERT baseline based on BERTScore (P, R, F1), response similarity and RAGAS context precision.

Retrieval Strategy	Query Embeddings	Context Embeddings	Output Model	BERTScore P	BERTScore R	BERTScore F1	Exact Match (EM)	F1 Score (Token-Based)	Response Similarity (Max Value)	RAGAS Context Precision
Baseline	-	-	Fine-Tuned mBERT	75.30%	78.10%	76.60%	45.10%	61.30%	-	-
Hybrid Retrieval (RRF)	text-embedding-3-small	text-embedding-3-small + BM25	GPT-4o	91.80%	94.57%	93.15%	78.45%	88.90%	97.80%	95.20%
Pure Semantic	text-embedding-3-small	Sentence Transformer : 'paraphrase-MiniLM-L6-v2'	GPT-4o	90.44%	93.04%	91.72%	71.20%	84.75%	96.84%	85.10%
Pure Semantic	text-embedding-3-small	text-embedding-3-small	GPT-4o	90.41%	93.46%	91.91%	70.95%	84.50%	63.35%	81.45%
Pure Semantic	text-embedding-3-small	Sentence Transformer : 'paraphrase-MiniLM-L6-v2'	Llama3.2-latest (3B)	87.50%	90.24%	88.85%	65.50%	82.10%	97.27%	85.10%
Pure Semantic	text-embedding-3-small	text-embedding-3-small	Llama3.2-latest (3B)	90.77%	92.06%	91.41%	68.20%	83.95%	55.88%	81.45%
Pure Semantic	text-embedding-3-small	Sentence Transformer : 'paraphrase-MiniLM-L6-v2'	Deepseek-r1	81.28%	88.82%	84.88%	58.70%	70.15%	12.77%	85.10%
Pure Semantic	text-embedding-3-small	text-embedding-3-small	Deepseek-r1	81.47%	88.26%	84.73%	57.90%	69.80%	47.61%	81.45%

Top-5 retrieval than the text-embedding-3-large model due to the ability to retrieve multiple relevant context passages. Text-embedding-3-small is more suitable in applications that demand a high coverage of contexts as it yields more results to the question answering systems that require supporting evidences. The text-embedding-3-large model proved to be more accurate in choosing the most appropriate match in Top-1 retrieval tasks hence suited in direct answer retrieval applications. The average similarity scores indicate that larger models provide only small benefits in best match retrieval but smaller models remain efficient and reliable in multi-document queries that is essential to information-dense QA systems.

Furthermore, the evaluation of LLM-generated responses demonstrated that GPT-4o consistently delivered the highest performance across all key metrics, including Exact Match (78.45%), F1 Score (88.90%), precision, recall, and response similarity. This indicates that GPT-4o is particularly well-suited for generating high-quality, factually precise responses when paired with effective retrieval mechanisms. Llama-3.2 (3B) exhibited strong performance, particularly in response similarity, but still fell short of GPT-4o in

Exact Match and F1 Score measurements, suggesting that while it can produce semantically relevant responses, it may lack fine-grained accuracy in matching user queries to retrieved contexts. Deepseek-r1, despite achieving reasonable context-matching metrics, struggled significantly in generating contextually relevant responses, as evident from its low response similarity scores and corresponding lower EM and F1 values. This discrepancy highlights the challenge of selecting LLMs that are not only capable of interpreting embeddings but also adept at synthesizing coherent and accurate responses.

This work shows that the retrieval strategy and LLM choice are a crucial component of optimising Urdu QA systems based on Retrieval-Augmented Generation (RAG). The text-embedding-3-small model was better at Top-3 and Top-5 retrieval than the text-embedding-3-large model due to the ability to retrieve multiple relevant context passages. Text-embedding-3-small is more suitable in applications that demand a high coverage of contexts as it yields more results to the question answering systems that require supporting evidences. The text-embedding-3-large model proved to be more accurate in choosing the most appropriate match in

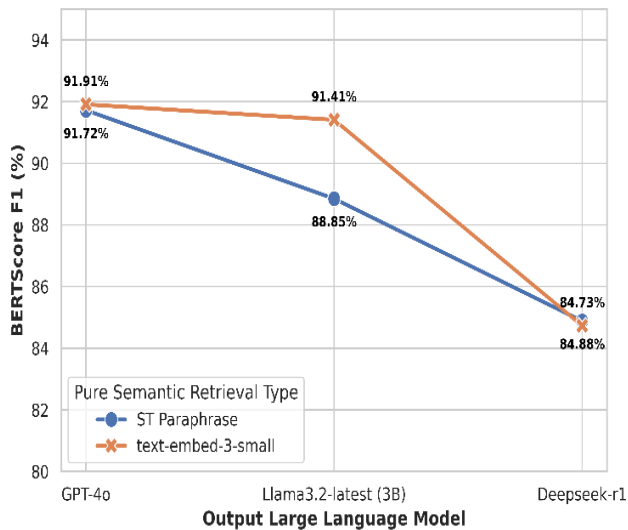


FIGURE 3. Sensitivity analysis of BERTScore F1 performance across different output LLMs and pure semantic context retrieval types.

Top-1 retrieval tasks hence suited in direct answer retrieval applications. The average similarity scores indicate that larger models provide only small benefits in best match retrieval but smaller models remain efficient and reliable in multi-document queries that is essential to information-dense QA systems.

Furthermore, the evaluation of LLM-generated responses demonstrated that GPT-4o consistently delivered the highest performance across all key metrics, including Exact Match (78.45%), F1 Score (88.90%), precision, recall, and response similarity. This indicates that GPT-4o is particularly well-suited for generating high-quality, factually precise responses when paired with effective retrieval mechanisms. Llama-3.2 (3B) exhibited strong performance, particularly in response similarity, but still fell short of GPT-4o in Exact Match and F1 Score measurements, suggesting that while it can produce semantically relevant responses, it may lack fine-grained accuracy in matching user queries to retrieved contexts. Deepseek-r1, despite achieving reasonable context-matching metrics, struggled significantly in generating contextually relevant responses, as evident from its low response similarity scores and corresponding lower EM and F1 values. This discrepancy highlights the challenge of selecting LLMs that are not only capable of interpreting embeddings but also adept at synthesizing coherent and accurate responses.

Although the UQA-RAG system can demonstrate good results in general, a qualitative examination of its failures and drawbacks is necessary to recognize the actual potential and shortcomings of the system. Our analysis of erroneous responses showed that there are a few typical failure scenarios, the most frequent of which was the recall of irrelevant or poorly connected context. An example of this would be a search on the topic of the functions of the National Assembly which would yield documents on the topic of parliamentary committees without delivering the high level functions. Cru-

cially, the implementation of the Hybrid Retrieval Strategy substantially reduced the failure rate due to poorly connected context by introducing lexical exactness to the ranking, which is reflected in the high Context Precision score (95.20%) and the superior Exact Match score (78.45%) for the Hybrid configuration. In some cases, the LLMs, especially Deepseek-r1 and to a lesser degree Llama-3.2, produced information that did not exist in the retrieved context. This typically happened when the given context was either excessively brief or not directly pertinent to the question that the user was asking, so the model would generate an answer that was plausible but wrong. The system occasionally had trouble with multi-hop queries that entailed synthesizing information across multiple, non-contiguous documents. As in the case of a query such as: “Who was the speaker of the National assembly when a certain bill was enacted?” may need to access information in two articles—one on the bill and the other on the speaker then. The Top-k method of the retrieval system was not always able to find all the required information, which resulted in either an incomplete or an incorrect final answer. Too general or ambiguous queries, tended to lead to very generic or divided answers. The retrieval component of the system would retrieve a broad set of documents, and the LLM could not easily combine them into one, coherent, and concise response. These weaknesses indicate that although the RAG method is strong, its effectiveness largely depends on the effectiveness of the retrieval stage. The next steps in the research should be the enhancement of the retrieval mechanism to support complex, multi-hop queries and the guarantee that the context retrieved is highly relevant to the specific intent of the query.

The development of UQA-RAG holds significant ethical and societal value, primarily by advancing technology in a low-resource language domain. By providing accurate, contextually-grounded information in Urdu, the system promotes digital inclusion and access to knowledge for millions of native Urdu speakers. However, the ethical deployment of this technology requires adherence to responsible AI principles. Specifically, the high RAGAS Context Precision achieved by the Hybrid Retrieval method is vital for mitigating the risk of spreading misinformation (hallucinations), which is a major ethical concern with general-purpose LLMs. Future work must focus on continuous monitoring for biases in the underlying UNQA dataset and ensuring that the system’s output remains neutral and factually correct. The open-source Llama-3.2 results provide a crucial pathway for the broader community to deploy a high-performing Urdu QA system independently, fostering technological self-reliance in this linguistic group.

The proven strength of the UQA-RAG framework, especially its high-accurate hybrid retrieval, creates an important opportunity to be used practically in other highly information-intensive industries. A case in point is in the educational field where the system may be modified to consume digitized textbooks and academic publication collections, giving the student a dynamic and convenient

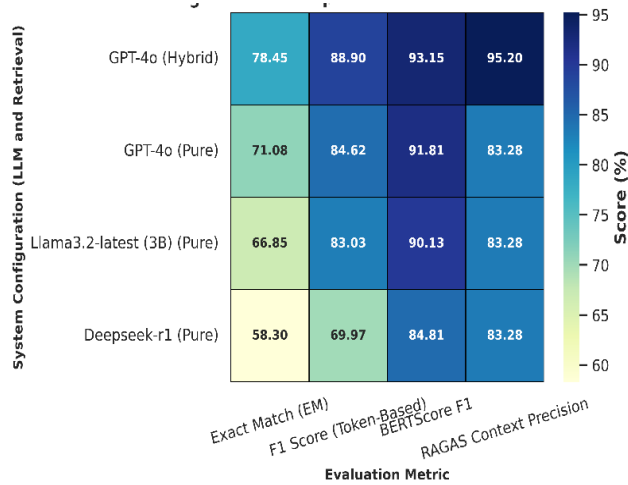


FIGURE 4. Comprehensive heatmap of all key RAG configurations across core and RAGAS evaluation metrics.

learning and research resource. In the same way within government services, it may be used to make the government more transparent by enabling citizens to query all parliamentary archives, public records or even the official gazettes with natural language queries. Moreover, the structure of the framework would fit well into the legal system, as it may help legal practitioners navigate through a dense body of case law and statutes in Urdu, helping to enhance the effectiveness of legal research and access to justice.

Embeddings of high fidelity coupled with suitable LLMs can be the key ingredients to optimizing QA system performance in languages like Urdu and others in the low-resource category. Users are advised to make a choice between text-embedding-3-small and text-embedding-3-large depending on their task needs to achieve a great context or exact accuracy. The Hybrid Retrieval approach provides the best overall balance of speed, accuracy, and context quality, validated by the high factual precision (EM 78.45%, F1 88.90%) and the 95.20% Context Precision score. GPT-4o proves to be most reliable in providing accurate context-based responses but Llama-3.2 (3B) provides other alternatives that can be suitable to specific applications. The results of the research can be used to create more effective Urdu QA systems since they demonstrate that users require both a reasonable retrieval speed and high-quality generative results to increase their satisfaction with reliable responses.

Despite the high quantitative performance of the UQA-RAG framework, a qualitative assessment of failure cases reveals certain system limitations. The retrieval component occasionally encounters challenges with semantic ambiguity, where linguistically similar but contextually unrelated information is prioritized. Furthermore, long-form and complex queries sometimes result in fragmented context retrieval, which can limit the model's ability to synthesize a complete response. Additionally, the handling of specialized terminology and script-specific variations remains a challenge for both lexical and semantic search layers. These observations suggest that while the hybrid approach significantly improves

accuracy, the system remains sensitive to query complexity and the nuances of low-resource linguistic processing.

VI. CONCLUSION

The research successfully examined how OpenAI embedding models and Large Language Models (LLMs) function together in UQA-RAG, a novel Retrieval-Augmented Generation (RAG) system, to boost Urdu Question Answering (QA) performance. The core findings validate the efficacy of the proposed Hybrid Retrieval Strategy (Semantic Search + BM25 + RRF), which achieved the highest overall RAGAS Context Precision score of 95.20%, confirming that fusing lexical and semantic signals significantly improves context quality compared to pure vector search methods. This methodological advancement led to the UQA-RAG system, specifically the Hybrid Retrieval Strategy combined with GPT-4o, demonstrating decisive superiority over the baseline. The system achieved the highest overall BERTScore F1 of 93.15%, and, critically, reached an Exact Match (EM) score of 78.45% and a token-based F1 Score of 88.90%. This represents a substantial improvement in factual precision and token overlap compared to the fine-tuned mBERT extractive baseline.

Further discussion revealed that the text-embedding-3-small model ranked higher in terms of Top-3 and Top-5 retrieval whereas the text-embedding-3-large model ranked higher in terms of Top-1 retrieval indicating that the best embedding model (selected) should be related to the desired retrieval task (Top-3/Top-5). GPT-4o appeared to be the best or superior of all the generative models, as it had the highest measure of accuracy, recall, and similarity of the response, so it can be the most reliable when it comes to generating quality responses. Llama-3.2 (3B) had good response similarity but lower precision and recall than GPT-4o and Deepseek-r1 was poor at contextual relevance even when its context-matching results were reasonable.

The development of UQA-RAG holds significant ethical and societal value by advancing technology in a low-resource language domain; the validated Hybrid Retrieval Strategy's high Context Precision is vital for mitigating the risk of LLM hallucination and promoting responsible information access for the Urdu community. Task requirements determine why researchers must choose suitable embedding models and LLMs for their work. Beyond its application to Urdu, the modular architecture of the UQA-RAG framework offers significant potential for generalizability to other low-resource and morphologically rich languages. The hybrid retrieval strategy balancing semantic context with lexical precision provides a scalable template that can be adapted to various linguistic environments by updating the language specific embedding models and preprocessing pipelines. Future research should focus on enhancing the retrieval mechanism to support complex multi-hop queries, optimizing the open-source Llama-3.2 (3B) model for Urdu grounding to provide a fully accessible solution, and continuing to explore hybrid approaches that integrate multiple embeddings and generative models to further enhance performance. By addressing

challenges in low-resource language processing, this research lays the foundation for more scalable and effective AI-driven QA solutions in Urdu and establishes a foundational methodology for enhancing digital information access across a broader spectrum of under-resourced languages.

ACKNOWLEDGMENT

The Researchers would like to thank the Deanship of Graduate Studies and Scientific Research at Qassim University for financial support (QU-APC-2026).

REFERENCES

- [1] F. Yang, P. Zhao, Z. Wang, L. Wang, J. Zhang, M. Garg, Q. Lin, S. Rajmohan, and D. Zhang, "Empower large language model to perform better on industrial domain-specific question answering," 2023, *arXiv:2305.11541*.
- [2] O. Ovadia, M. Brief, M. Mishaeli, and O. Elisha, "Fine-tuning or retrieval? Comparing knowledge injection in LLMs," 2023, *arXiv:2312.05934*.
- [3] N. Kandpal, H. Deng, A. Roberts, E. Wallace, and C. Raffel, "Large language models struggle to learn long-tail knowledge," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 15696–15707.
- [4] B. Saha and U. Saha, "Enhancing international graduate student experience through AI-driven support systems: A LLM and RAG-based approach," in *Proc. Int. Conf. Data Sci. Its Appl. (ICoDSA)*, Jul. 2024, pp. 300–304.
- [5] T. F. Khan, W. Anwar, H. Arshad, and S. N. Abbas, "An empirical study on authorship verification for low resource language using hyper-tuned CNN approach," *IEEE Access*, vol. 11, pp. 80403–80415, 2023, doi: [10.1109/ACCESS.2023.3299565](https://doi.org/10.1109/ACCESS.2023.3299565).
- [6] T. F. Khan, M. Sabir, M. H. Malik, H. Ghous, H. M. Ijaz, A. Nadeem, and A. Ejaz, "Comparative analysis of hybrid ensemble algorithms for authorship attribution in Urdu text," *J. Comput. Biomed. Inform.*, vol. 2024, pp. 1–15, Apr. 2024.
- [7] T. B. Brown et al., "Language models are few-shot learners," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, Jan. 2020, pp. 1877–1901.
- [8] S. Chen, Y. Hou, Y. Cui, W. Che, T. Liu, and X. Yu, "Recall and learn: Fine-tuning deep pretrained language models with less forgetting," 2020, *arXiv:2004.12651*.
- [9] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners," *OpenAI Blog*, vol. 1, no. 8, p. 9, 2019.
- [10] A. Chowdhery et al., "PaLM: Scaling language modeling with pathways," *J. Mach. Learn. Res.*, vol. 24, no. 240, pp. 1–113, 2022.
- [11] H. Touvron et al., "Llama 2: Open foundation and fine-tuned chat models," 2023, *arXiv:2307.09288*.
- [12] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, E. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2025, pp. 1–11.
- [13] M. U. Hadi, Q. A. Tashi, R. Qureshi, A. Shah, A. Muneer, M. Irfan, A. Zafar, M. B. Shaikh, N. Akhtar, J. Wu, and S. Mirjalili, "A survey on large language models: Applications, challenges, limitations, and practical usage," *Authorea Preprints*, vol. 2023, pp. 1–31, Jan. 2023.
- [14] S. Huo, N. Arabzadeh, and C. L. A. Clarke, "Retrieving supporting evidence for LLMs generated answers," 2023, *arXiv:2306.13781*.
- [15] L. Fan, L. Li, Z. Ma, S. Lee, H. Yu, and L. Hemphill, "A bibliometric review of large language models research from 2017 to 2023," 2023, *arXiv:2304.02020*.
- [16] V. Bolotova, V. Blinov, F. Scholer, W. B. Croft, and M. Sanderson, "A non-factoid question-answering taxonomy," in *Proc. 45th Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, Jul. 2022, pp. 1196–1207.
- [17] A. Rogers, M. Gardner, and I. Augenstein, "QA dataset explosion: A taxonomy of NLP resources for question answering and reading comprehension," *ACM Comput. Surveys*, vol. 55, no. 10, pp. 1–45, Oct. 2023.
- [18] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Comput. Surveys*, vol. 55, no. 12, pp. 1–38, Dec. 2023.
- [19] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-T. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 9459–9474.
- [20] G. Mialon, R. Dessì, M. Lomeli, C. Nalmpantis, R. Pasunuru, R. Raileanu, B. Rozière, T. Schick, J. Dwivedi-Yu, A. Celikyilmaz, E. Grave, Y. LeCun, and T. Scialom, "Augmented language models: A survey," 2023, *arXiv:2302.07842*.
- [21] A. Lazaridou, E. Gribovskaya, W. Stokowiec, and N. Grigorev, "Internet-augmented language models through few-shot prompting for open-domain question answering," 2022, *arXiv:2203.05115*.
- [22] P. Lewis, Y. Wu, L. Liu, P. Minervini, H. Küttler, A. Piktus, P. Stenetorp, and S. Riedel, "PAQ: 65 million probably-asked questions and what you can do with them," *Trans. Assoc. for Comput. Linguistics*, vol. 9, pp. 1098–1115, Oct. 2021.
- [23] M. Park, H. Oh, E. Choi, and W. Hwang, "LRAGE: Legal retrieval augmented generation evaluation tool," 2023, *arXiv:2504.01840*.
- [24] M. Lee and Y. Cho, "Improving large language model applications in biomedicine with retrieval-augmented generation: A systematic review, meta-analysis, and clinical development guidelines," *J. Amer. Med. Inform. Assoc.*, vol. 32, no. 4, pp. 605–614, 2025.
- [25] S. Khoja and S. Ahmed, "UQAD1.0: Development of an Urdu question answering training data for machine reading comprehension," 2021, *arXiv:2111.01543*.
- [26] S. Arif, S. Farid, A. Athar, and A. A. Raza, "UQA: Corpus for Urdu question answering," 2024, *arXiv:2405.01458*.
- [27] F. Faissal and U. Yousaf, "LEGAL-UQA: A low-resource Urdu–English dataset for legal question answering," 2024, *arXiv:2410.13013*.
- [28] C. Sharma, "Retrieval-augmented generation: A comprehensive survey of architectures, enhancements, and robustness frontiers," 2025, *arXiv:2506.00054*.
- [29] V. Sharma and V. Raman, "A reliable knowledge processing framework for combustion science using foundation models," 2023, *arXiv:2401.00544*.
- [30] I. Siragusa and R. Pirrone, "Conditioning chat-GPT for information retrieval: The Unipa-GPT case study," in *Proc. 7th Workshop Natural Lang. Artif. Intell. (NLAAI) Co-Located 22th Int. Conf. Italian Assoc. Artif. Intell. (AI IA)*, 2023, pp. 1–11.
- [31] M. Kulkarni, P. Tangarajan, K. Kim, and A. Trivedi, "Reinforcement learning for optimizing rag for domain chatbots," 2024, *arXiv:2401.06800*.
- [32] S. Campese, I. Lauriola, and A. Moschitti, "QUADRo: Dataset and models for QuesTion-answer database retrieval," 2023, *arXiv:2304.01003*.
- [33] B. Saha, U. Saha, and M. Z. Malik, "QuIM-RAG: Advancing retrieval-augmented generation with inverted question matching for enhanced QA performance," *IEEE Access*, vol. 12, pp. 185401–185410, 2024, doi: [10.1109/ACCESS.2024.3513155](https://doi.org/10.1109/ACCESS.2024.3513155).
- [34] A. Mahmood, Z. Ahmad, I. Ameer, and G. Sidorov, "Information retrieval vs cache augmented generation vs fine tuning: A comparative study on Urdu medical question answering," in *Proc. CS & IT Conf.*, vol. 15, no. 10, 2025.
- [35] N. Muennighoff, N. Tazi, L. Magne, and N. Reimers, "MTEB: Massive text embedding benchmark," 2022, *arXiv:2210.07316*.
- [36] S. Kumar, A. Solanki, and N. Z. Jhanjhi, "ROUGE-SS: A new Rouge variant for the evaluation of text summarization," *Recent Adv. Comput. Sci. Commun.*, vol. 18, no. 6, 2025, Art. no. E060624230748.
- [37] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, "BERTScore: Evaluating text generation with BERT," 2019, *arXiv:1904.09675*.
- [38] E. S. Shahul, J. James, L. Espinosa-Anke, and S. Schockaert, "RAGAS: Automated evaluation of retrieval augmented generation," 2023, *arXiv:2309.15217*.



TALHA FAROOQ KHAN received the Ph.D. degree in computer science from The Islamia University of Bahawalpur, Pakistan. He is currently an Assistant Professor with the Department of Computer Science, The University of Faisalabad, Pakistan. He has more than nine years of experience in teaching and research. He has published several research papers in well-reputed international journals and conference proceedings. He also serves as a reviewer for various peer-reviewed journals and has contributed to multiple research collaborations. His primary research interests include natural language processing, text mining, web mining, machine learning, deep learning, and LLMs.



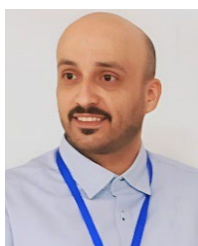
contribute meaningfully to the scientific community and pave the way for novel applications in AI and data science.

MUHAMMAD ARSLAN is currently pursuing the M.Phil. degree in computer science with the University of Southern Punjab, Multan. Leveraging his strong academic background, he delves into the intricacies of artificial intelligence, data science, and machine learning. He is a Driven Researcher. His research endeavors focus on exploring the frontiers of deep learning and generative AI, with a keen eye on harnessing these technologies to drive innovation. Through his work, he aims to



Saudi Arabia. His research interests include wireless networks, machine learning, and deep learning.

AHMED H. ALMUTAIRI received the B.S. degree in computer science from the Department of Computer Science, Western Oregon University, Monmouth, OR, USA, in 2017, the M.S. degree in computer science from Southern Illinois University, Carbondale, IL, USA, in 2018, and the Ph.D. degree in computer science from Portland State University, OR, in 2024. Currently, he is an Assistant Professor with the Department of Computer Science, Northern Border University,



member of the Australian Computer Society (ACS) for four years and served as a reviewer and a technical program committee (TPC) member for numerous journals and conferences.

FAHAD ALTURISE received the Ph.D. degree in information technology from Flinders University. He is currently a Professor with the Department of Cybersecurity, College of Computer, Qassim University. He has 16 years of experience in teaching and research. He has published more than 70 papers in international journals and conference proceedings. His primary research interests include e-services, AI, ML, IOT, ICT adaptation, IT security, and software engineering. He was a



to the field through research, development, and collaboration.

MUHAMMAD TARIQ RAZA is currently pursuing the degree in computer science with Bahauddin Zakariya University, Multan. Passionate about technology and innovation, he has a keen interest in areas, such as artificial intelligence, data science, and software development. With a strong academic foundation, he continuously explores emerging trends in the tech industry, particularly in AI, machine learning, and automation. He is dedicated to expanding his expertise and contributing

...